Math 55—Fall 2025—Haiman
Homework 3
Due on Gradescope Wednesday, Sept. 17, 8pm

Problems from Rosen

Section 2.3

12 (c,d) (justify your answers).
14 (b,d) (justify your answers).
34.
40.
50.

Section 2.5

2 (c,d,e,f).
10 (and explain).

Additional problem

In class on Mon. 9/15, we'll construct a bijection between $\mathbb{N}$ and $\mathbb{N}\times\mathbb{N}$ by counting along diagonals, as indicated by this diagram:

$$\begin{array}{ccccccc} 0 & 1 & 3 & 6 & 10 & & \\ 2 & 4 & 7 & 11 & & & \\ 5 & 8 & 12 & & \dots & \longleftrightarrow & \\ 9 & 13 & & & & & \\ 14 & & & & & & \end{array} \quad \begin{array}{ccccccc} (0,0) & (1,0) & (2,0) & (3,0) & (4,0) & \\ (0,1) & (1,1) & (2,1) & (3,1) & & \\ (0,2) & (1,2) & (2,2) & & \dots & . \\ (0,3) & (1,3) & & & & \\ (0,4) & & & & & \end{array}$$

Show that the function $f\colon \mathbb{N}\times\mathbb{N}\to\mathbb{N}$ described by the diagram is given by the formula

$$f((i,j)) = (i+j)(i+j+1)/2 + j\,.$$

(Once you verify this, the diagram shows that the formula defines a bijective function. It is also possible to prove the latter fact algebraically, without referring to the diagram—see the answer to Rosen 2.5 Ex. 31 for the proof of a similar result with $\mathbb{Z}^+$ in place of $\mathbb{N}$.)